Data Communications & Networks

Chapter 5: Sections 5.1 & 5.2

Analog Transmission

Digital-to-Analog & Analog-to-Analog Conversion

Topics Covered:

- 5.1.1 Aspects: Bit rate, Baud rate, Carrier signal
- 5.1.2 Amplitude Shift Keying (ASK / BASK / OOK)
- 5.1.3 Frequency Shift Keying (FSK / BFSK / MFSK)
- 5.1.4 Phase Shift Keying (PSK / BPSK / QPSK)
- 5.1.5 Quadrature Amplitude Modulation (QAM)
- 5.2.1 Amplitude Modulation (AM)
- 5.2.2 Frequency Modulation (FM)
- 5.2.3 Phase Modulation (PM)

Reference: Forouzan, *Data Communications and Networking*, 5th ed.

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Chapter 1 Digital-to-Analog Conversion — Section 5.1

The Big Picture: Why Analog Transmission of Digital Data?

Digital signals are great inside a computer, but the real world is mostly **bandpass**. A bandpass channel can only carry signals within a specific frequency range (e.g., a telephone line passes 300–3400 Hz). A raw digital pulse has energy from DC (0 Hz) all the way up — it cannot pass through a bandpass channel.
The solution: Convert digital data into an analog sine wave whose frequency fits within the available bandpass channel. The original sine wave (the **carrier**) is modified to carry the digital information.
We alter one or more of the three properties of a sine wave:

Property altered	Technique	Abbreviation	Noise resistance
Amplitude only	Amplitude Shift Keying	ASK	Lowest
Frequency only	Frequency Shift Keying	FSK	Medium
Phase only	Phase Shift Keying	PSK	Good
Amplitude + Phase	Quadrature Amplitude Mod.	QAM	Best

1.1 Aspects of Digital-to-Analog Conversion — 5.1.1

1.1.1 Bit Rate vs. Baud Rate

Two Rates — One Link

- **Bit rate (N):** number of *bits* sent per second (bps). This is what the application cares about.
- **Baud rate (S):** number of *signal elements* sent per second. This is what the channel "sees." Signal rate = baud rate.

Each signal element can carry r bits, where $r = \log_2 L$ and L = number of distinct signal levels.

Baud Rate Formula

$$S = N \times \frac{1}{r} \quad \Longleftrightarrow \quad N = S \times r \qquad \text{where } r = \log_2 L$$

Key rule: In analog transmission, baud rate $\leq$ bit rate (always). More levels per signal element $\Rightarrow$ fewer signal changes per second $\Rightarrow$ less bandwidth needed.

Analogy — Cars and Passengers: Think of a baud as a *car* and a bit as a *passenger*. To move 1000 people (bits/s) you can either send 1000 single-occupancy cars (1 bit/baud), or 500 two-seater cars (2 bits/baud), or 250 four-seater cars (4 bits/baud). Same road capacity, more passengers per trip.

Example 5.1 — Finding Bit Rate from Baud Rate

Given: An analog signal carries 4 bits per signal element. 1000 signal elements are sent per second. Find the bit rate.

Solution:

$$r = 4, \quad S = 1000 \text{ baud}, \quad N = ?$$

$$N = S \times r = 1000 \times 4 = \mathbf{4000} \text{ bps}$$

Example 5.2 — Finding Number of Levels from Bit and Baud Rate

Given: Bit rate = 8000 bps, baud rate = 1000 baud. Find r and L.

Solution:

$$S = N \times \frac{1}{r} \;\Rightarrow\; r = \frac{N}{S} = \frac{8000}{1000} = 8 \text{ bits/baud}$$

$$r = \log_2 L \;\Rightarrow\; L = 2^r = 2^8 = \mathbf{256} \text{ signal levels}$$

Each signal element encodes 8 bits and there are 256 distinct waveform types.

1.1.2 Carrier Signal

What Is a Carrier Signal?

The carrier is a simple, high-frequency sine wave:

$$s(t) = A\cos(2\pi f_c t + \phi)$$

- A = peak amplitude (Volts)
- f_c = carrier frequency (Hz) — chosen to match the available bandpass channel
- ϕ = initial phase (degrees or radians)

Digital information is encoded by *changing* A, f_c, or ϕ. The process of changing the carrier is called **modulation** (or *shift keying* when the change is discrete).

1.2 Amplitude Shift Keying (ASK) — 5.1.2

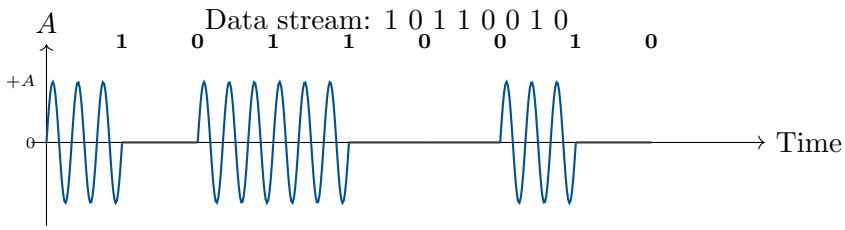
1.2.1 Binary ASK (BASK / OOK)

How BASK Works

- **Binary 1** → carrier at full amplitude A.
- **Binary 0** → carrier at zero amplitude (silent).

Because one level is "on" and the other is "off", BASK is also called **On-Off Keying (OOK)**.

Frequency and phase of the carrier remain *constant*; only amplitude changes.



BASK (OOK): carrier ON for 1, carrier OFF for 0

1.2.2 Bandwidth for ASK

ASK Bandwidth Formula

$$B_{\text{ASK}} = (1 + d) \times S$$

- S = baud rate (signal elements/second)
- d = factor depending on modulation/filtering, $0 \leq d \leq 1$
- Minimum bandwidth (d=0): $B = S$
- Maximum bandwidth (d=1): $B = 2S$
- Carrier frequency f_c sits at the *centre* of the bandwidth.

Why is f_c centred? The modulation process creates sidebands above *and* below f_c. By choosing f_c to be the midpoint of the available bandpass channel, we make full use of the channel and avoid leaking into other channels.

Example 5.3 — ASK over a 100-kHz bandpass channel ($d = 1$)

Given: Available bandwidth = 100 kHz spanning 200–300 kHz; $d = 1$. Find carrier frequency and maximum bit rate.

Solution:

$$f_c = \frac{200 + 300}{2} = \mathbf{250}\ \text{kHz} \quad \text{(midpoint)}$$

With $d = 1$ and $r = 1$ (binary):

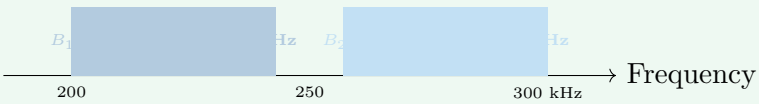
$$B = (1 + d)\,S \;\Rightarrow\; 100 = 2S \;\Rightarrow\; S = 50 \text{ kbaud}$$

$$N = S \times r = 50 \times 1 = \mathbf{50} \text{ kbps}$$

Example 5.4 — Full-Duplex ASK (two directions)

Given: Same 100-kHz channel (200–300 kHz), full-duplex with two sub-bands. Find bit rate per direction.

Solution: Split the 100 kHz into two 50-kHz sub-bands:



Each direction has $B = 50$ kHz and, with $d = 1$:

$$50 = 2N \;\Rightarrow\; N = \mathbf{25} \text{ kbps per direction}$$

1.2.3 Noise Weakness of ASK

ASK is the **most noise-susceptible** of all four methods. Noise directly adds to or subtracts from the amplitude, the very quantity being measured. Even small noise can flip a "1" (high amplitude) to look like a "0" (zero amplitude) or vice versa.

1.3 Frequency Shift Keying (FSK) — 5.1.3

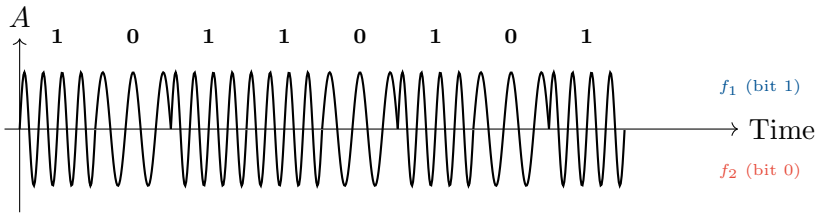
1.3.1 Binary FSK (BFSK)

How BFSK Works

- **Binary 1** → carrier at frequency f_1 (higher).
- **Binary 0** → carrier at frequency f_2 (lower).

Amplitude and phase remain *constant*; only frequency jumps between f_1 and f_2.

The separation between the two carrier frequencies is $2\Delta_f$.



BFSK: higher frequency for 1, lower frequency for 0

1.3.2 Bandwidth for BFSK

BFSK Bandwidth Formula

$$B_{\text{BFSK}} = (1 + d)\, S + 2\Delta_f$$

- The extra term $2\Delta_f$ accounts for the gap between the two carrier frequencies.
- Minimum required separation: $2\Delta_f \geq S$ (to avoid overlap of the two sub-bands).

1.3.3 Multilevel FSK (MFSK)

Instead of two frequencies, use L different frequencies, sending $\log_2 L$ bits per baud.

MFSK Bandwidth Formula

$$B_{\text{MFSK}} = L \times S$$

(assuming minimum separation $2\Delta_f = S$ between adjacent carriers)

Worked Examples

Example 5.5 — BFSK over a 100-kHz channel ($d = 1$)

Given: Bandwidth = 100 kHz (200–300 kHz); $d = 1$. Choose $2\Delta_f = 50$ kHz. Find S and bit rate.

Solution:

$$B = (1 + d)\, S + 2\Delta_f \;\Rightarrow\; 100 = 2S + 50 \;\Rightarrow\; S = 25 \text{ kbaud}$$

$$N = S \times r = 25 \times 1 = \mathbf{25} \text{ kbps}$$

Comparison with Example 5.3 (ASK): ASK gives 50 kbps on the same channel; FSK gives only 25 kbps — FSK is less spectrally efficient but more noise-resistant.

Example 5.6 — MFSK: 3 bits per baud at 3 Mbps

Given: Send 3 bits per baud; bit rate = 3 Mbps; carrier = 10 MHz. Find L, baud rate, and bandwidth.

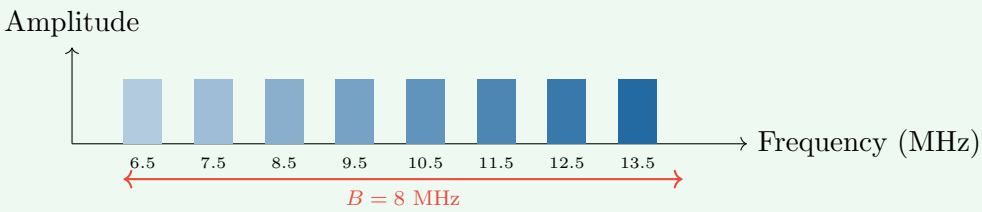
Solution:

$$L = 2^3 = \mathbf{8} \text{ frequencies}$$

$$S = \frac{N}{r} = \frac{3 \text{ Mbps}}{3} = \mathbf{1} \text{ Mbaud}$$

$$2\Delta_f = S = 1 \text{ MHz} \quad \text{(minimum separation)}$$

$$B = L \times S = 8 \times 1 = \mathbf{8} \text{ MHz}$$



Carrier $f_c = 10$ MHz centred; 8 frequencies at 1 MHz apart

1.4 Phase Shift Keying (PSK) — 5.1.4

1.4.1 Binary PSK (BPSK)

How BPSK Works

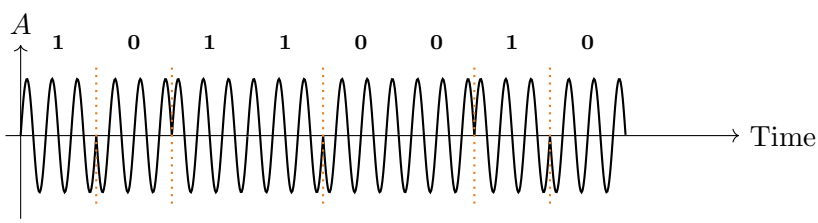
- **Binary 1** → carrier phase = 0 (normal sine wave)
- **Binary 0** → carrier phase = 180 (inverted sine wave)

Amplitude and frequency remain constant; only phase flips by 180°.

Why better than ASK?

- Noise affects *amplitude* far more easily than *phase*.
- PSK therefore has better Bit Error Rate (BER) than ASK under the same noise conditions.

Bandwidth: Same formula as ASK: $B = (1 + d)\, S$.



BPSK: same amplitude, phase flips 180° for each 0↔1 transition

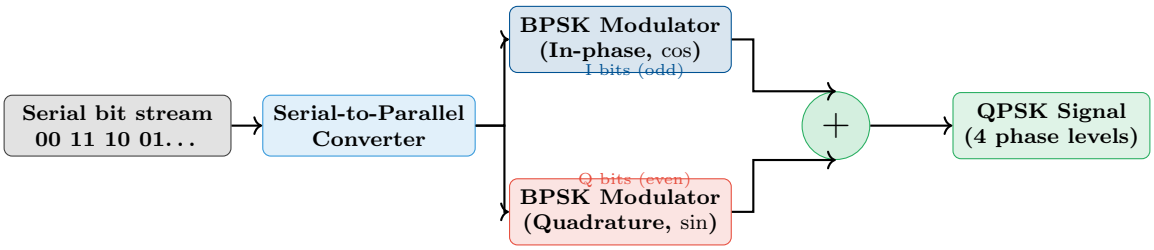
1.4.2 Quadrature PSK (QPSK)

QPSK: Two Bits per Baud

QPSK sends **2 bits** per signal element by using **four phase values**: 45, 135, −135, −45 (equivalently 45, 135, 225, 315).
How it works:

1. Incoming bitstream is split into odd bits (I channel) and even bits (Q channel).
2. Each channel runs its own BPSK modulator.
3. The I modulator uses a cosine carrier; the Q modulator uses a sine carrier (90° offset).
4. The two BPSK outputs are *added* together, producing four possible combined phases.

Result: $r = 2$, so $S = N/2$ (half the baud rate of BPSK for the same bit rate). This **halves the required bandwidth**!



Example 5.7 — QPSK Bandwidth at 12 Mbps ($d = 0$)

Given: Bit rate $N = 12$ Mbps; modulation = QPSK ($r = 2$); $d = 0$.

Solution:

$$S = \frac{N}{r} = \frac{12}{2} = 6 \text{ Mbaud}$$

$$B = (1 + 0) \times 6 = \mathbf{6} \text{ MHz}$$

Compare with BPSK at same 12 Mbps: $S = 12$ Mbaud $\Rightarrow B = 12$ MHz. QPSK uses **half the bandwidth** of BPSK for the same bit rate!

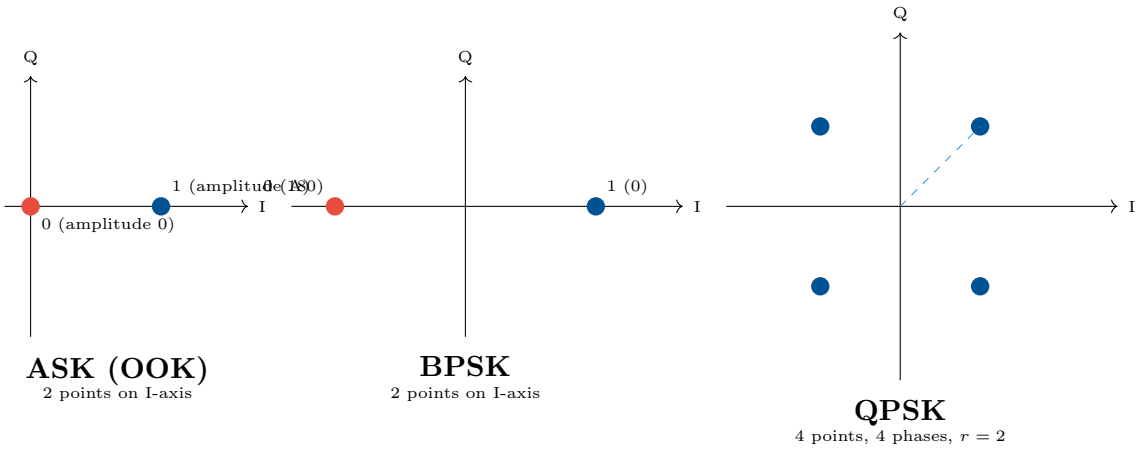
1.4.3 Constellation Diagram

Reading a Constellation Diagram

A constellation diagram is a 2-D plot where:

- **X-axis (I):** In-phase component amplitude.
- **Y-axis (Q):** Quadrature (90° offset) component amplitude.
- Each **dot** represents one distinct signal element.
- **Distance from origin** = peak amplitude of that signal element.
- **Angle from X-axis** = phase of that signal element.

The more dots a constellation has, the more bits per baud — but also the closer the dots are together, making them harder to tell apart in noisy conditions.



1.5 Quadrature Amplitude Modulation (QAM) — 5.1.5

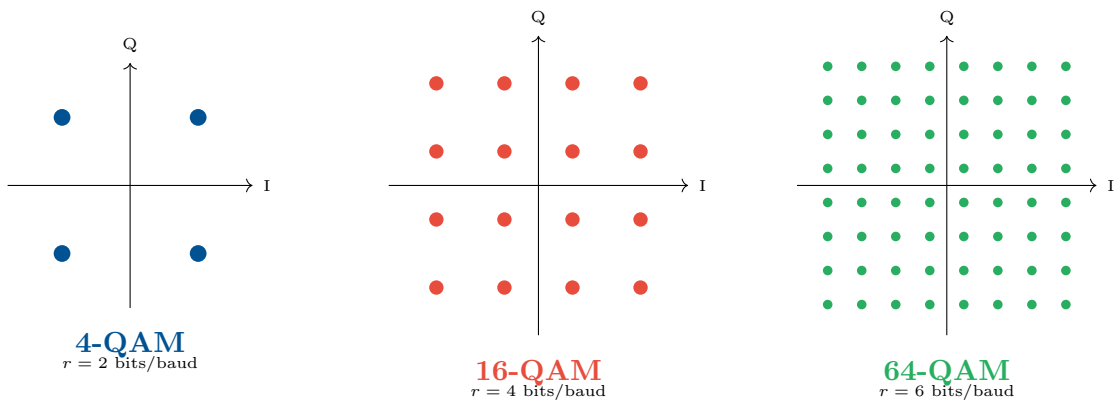
QAM: Changing Both Amplitude AND Phase

PSK is limited because beyond a certain number of phases, even tiny noise causes the receiver to mistake one phase for an adjacent one. The fix: also vary *amplitude*. By changing **both amplitude and phase simultaneously**, QAM can encode many more bits per baud.

$$\text{QAM} = \text{ASK} + \text{PSK}$$

- Uses two carriers: one in-phase (cosine) and one quadrature (sine).
- Each carrier is independently amplitude-modulated.
- Combined result has both amplitude and phase variation.
- Number of constellation points $L = 4, 8, 16, 32, 64, 128, 256, \ldots$
- Bits per baud: $r = \log_2 L$

1.5.1 QAM Variants and Constellation Diagrams



1.5.2 QAM Bandwidth and Numerical Examples

QAM Bandwidth and Bit Rate

Same bandwidth formula as ASK/PSK:

$$B_{\text{QAM}} = (1+d) \times S \qquad \text{where} \quad S = \frac{N}{r}, \quad r = \log_2 L$$

QAM achieves the same bandwidth as PSK but can encode *far* more bits per baud.

Problem P5.1d — 64-QAM Baud Rate

Given: Bit rate = 36,000 bps; modulation = 64-QAM. Find baud rate.

Solution:

$$L = 64 \;\Rightarrow\; r = \log_2 64 = 6 \text{ bits/baud}$$

$$S = \frac{N}{r} = \frac{36{,}000}{6} = \mathbf{6{,}000} \text{ baud}$$

Only 6000 signal changes per second are needed to carry 36 kbps!

Problem P5.6 — Bits per Baud from Constellation Size

Given: Find bits per baud for constellations of 2, 4, 16, 1024 points.

Points (L)	$r = \log_2 L$	**Example use**
2	1 bit/baud	BASK, BFSK, BPSK
4	2 bits/baud	QPSK, 4-QAM
16	4 bits/baud	16-QAM
1024	10 bits/baud	Advanced cable modems

Problem P5.7 — Bandwidth for 4000 bps ($d = 1$)

Find bandwidth for each modulation sending 4000 bps.

Method	r	S **(baud)**	B **(Hz)**
ASK ($d = 1$)	1	4000	$(1+1) \times 4000 = \mathbf{8000}$
FSK ($d = 1$, $2\Delta_f = 4$ kHz)	1	4000	$2 \times 4000 + 4000 = \mathbf{12{,}000}$
QPSK ($d = 1$)	2	2000	$2 \times 2000 = \mathbf{4000}$
16-QAM ($d = 1$)	4	1000	$2 \times 1000 = \mathbf{2000}$

Insight: Higher-order modulations (16-QAM) require far less bandwidth for the same bit rate — at the cost of needing a cleaner channel.

Problem P5.8 — Max Bit Rate on a 4-kHz Telephone Line ($d = 0$)

Given: Telephone bandwidth $B = 4$ kHz; $d = 0$.

With $d = 0$: $B = S \Rightarrow S = 4000$ baud. Then $N = S \times r = 4000 \times r$.

Method	r	Max bit rate
ASK (BASK)	1	$4000 \times 1 =$ **4** kbps
QPSK	2	$4000 \times 2 =$ **8** kbps
16-QAM	4	$4000 \times 4 =$ **16** kbps
64-QAM	6	$4000 \times 6 =$ **24** kbps

This illustrates why modern DSL modems use high-order QAM to squeeze megabits through the old telephone copper pair!

Problem P5.9 — QAM for 10 Channels of 10 Mbps each on 1-MHz medium ($d = 0$)

Given: Medium bandwidth = 1 MHz. Divide into 10 channels; each must carry 10 Mbps.

Solution: Each channel gets: $\frac{1 \text{ MHz}}{10} = 100$ kHz of bandwidth.
With $d = 0$: $B = S \Rightarrow S = 100$ kbaud per channel.
Required bits per baud:

$$r = \frac{N}{S} = \frac{10 \text{ Mbps}}{100 \text{ kbaud}} = \frac{10{,}000}{100} = 100 \text{ bits/baud}$$

$$L = 2^r = 2^{100} \approx 1.27 \times 10^{30} \text{ constellation points}$$

Conclusion: This is practically impossible! The Shannon limit for this channel (even with perfect SNR) would cap the data rate far below 10 Mbps. This problem illustrates that you cannot overcome the Nyquist/Shannon limits just by increasing L.

Problem P5.10 — 64-QAM on a 6-MHz Cable TV Channel

Given: Cable TV channel bandwidth = 6 MHz; modulation = 64-QAM; $d = 0$. Find available data rate per resident.

Solution:

$$r = \log_2 64 = 6 \text{ bits/baud}$$

With $d = 0$: $B = S \Rightarrow S = 6$ Mbaud.

$$N = S \times r = 6 \text{ Mbaud} \times 6 = \mathbf{36} \text{ Mbps}$$

This is why cable internet (DOCSIS standard) can deliver 30–50 Mbps per channel to subscribers.

1.5.3 Complete Comparison: ASK vs FSK vs PSK vs QAM

Property	**ASK**	**FSK**	**PSK**	**QAM**
What changes	Amplitude	Frequency	Phase	Amp + Phase
Bits per baud (r)	$\log_2 L$	$\log_2 L$	$\log_2 L$	$\log_2 L$
Bandwidth ($d = 0$)	S	$L \times S$	S	S
Noise resistance	Lowest	Medium	Good	Best (highest r)
Hardware complexity	Simplest	Moderate	Moderate	Most complex
Common use today	OOK (fibre)	Modems, radio	Wi-Fi, 5G	Cable, DSL, Wi-Fi

Chapter 2 Analog-to-Analog Conversion — Section 5.2

Why Modulate an Analog Signal?

An audio signal (voice, music) is already analog — why do we need to change it?
Problem: Every radio station produces a voice/music signal in the same low frequency range (roughly 0–15 kHz). If every station just broadcast raw audio, all signals would occupy the same frequencies and jam each other.
Solution: Each station *shifts* its audio signal to a unique, government-assigned carrier frequency using one of three analog modulation methods: AM, FM, or PM.

- The **modulating signal** = original audio (low-pass, low frequency).
- The **carrier signal** = high-frequency sine wave assigned to that station.
- The **modulated signal** = the carrier modified by the audio — what actually travels through the air.

2.1 Amplitude Modulation (AM) — 5.2.1

How AM Works

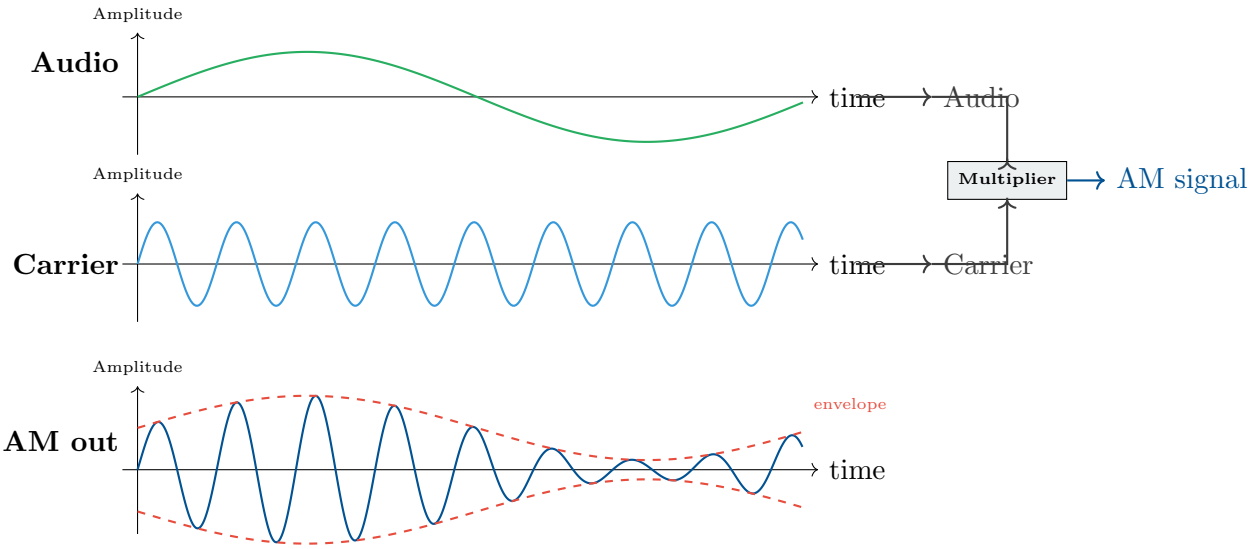
In AM:

- The **amplitude** of the carrier varies in step with the amplitude of the audio.
- Frequency and phase of the carrier remain *constant*.
- The audio signal becomes the *envelope* of the modulated carrier.

$$s(t) = \left[1 + k\,m(t)\right] A_c \cos(2\pi f_c t)$$

where $m(t)$ is the audio (modulating) signal and k is a modulation index.

Implementation: Multiply the audio signal by the carrier (a simple multiplier circuit).



2.1.1 AM Bandwidth

AM Bandwidth Formula

$$B_{\text{AM}} = 2B_{\text{audio}}$$

The modulation process creates two sidebands (upper and lower), each equal in bandwidth to the audio signal. Since both sidebands carry identical information, some systems use **Single Sideband (SSB)** AM and halve the bandwidth.

AM Radio Standard (Real World)

- Audio bandwidth: $B_{\text{audio}} = 5$ kHz (speech + music, AM quality).
- Required channel bandwidth: $B_{\text{AM}} = 2 \times 5 = \mathbf{10}$ kHz.
- FCC allocation: exactly **10 kHz per AM station**.
- Carrier frequency range: **530–1700 kHz**.
- Station separation: at least 10 kHz (one bandwidth).
- Number of possible AM channels: $\dfrac{1700 - 530}{10} = 117$ (some reserved for guard bands).

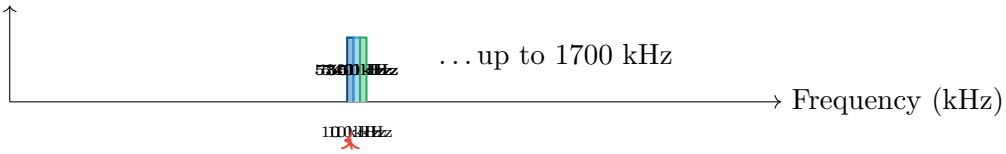
Problem P5.11a and P5.12a — AM Calculations

P5.11a: Find bandwidth for AM modulation of a 5-kHz voice signal.

$$B_{\text{AM}} = 2 \times 5 \text{ kHz} = \mathbf{10} \text{ kHz}$$

P5.12a: Total number of AM channels in the FCC-allocated band.

$$\text{Channels} = \frac{1700 - 530}{10} = \frac{1170}{10} = \mathbf{117} \text{ channels}$$



FCC AM Band: 530–1700 kHz, 10-kHz channels

2.2 Frequency Modulation (FM) — 5.2.2

How FM Works

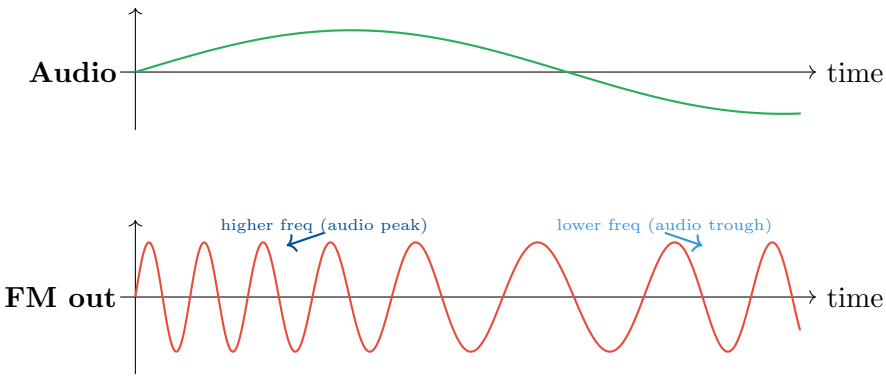
In FM:

- The **frequency** of the carrier varies in step with the *amplitude* of the audio.
- When the audio amplitude is high, the carrier frequency increases.
- When the audio amplitude is low (or negative), the carrier frequency decreases.
- Amplitude and initial phase of the carrier remain *constant*.

$$s(t) = A_c \cos\left(2\pi f_c t + 2\pi k_f \int_0^t m(\tau)\, d\tau\right)$$

where k_f is the frequency sensitivity constant.

Implementation: A **Voltage-Controlled Oscillator (VCO)** — the input voltage (audio level) controls the output frequency of the oscillator.



2.2.1 FM Bandwidth

FM Bandwidth Formula

$$B_{\text{FM}} = 2\,(1+\beta)\,B_{\text{audio}}$$

- β = modulation index (depends on how much the frequency deviates).
- Typical value: $\beta \approx 4$ for broadcast FM.
- Higher $\beta \Rightarrow$ wider bandwidth but better noise immunity (FM captures effect).

FM Radio Standard (Real World)

- Audio bandwidth: $B_{\text{audio}} = 15$ kHz (stereo broadcast quality).
- $\beta = 4$ (FCC standard for broadcast FM).
- Required channel bandwidth:

$$B_{\text{FM}} = 2(1+4) \times 15 = 2 \times 5 \times 15 = \mathbf{150} \text{ kHz}$$

- FCC allocation: **200 kHz per FM station** (150 kHz + guard band on each side).
- Carrier frequency range: **88–108 MHz**.
- Station separation: at least 200 kHz.

Problem P5.11b and P5.12b — FM Calculations

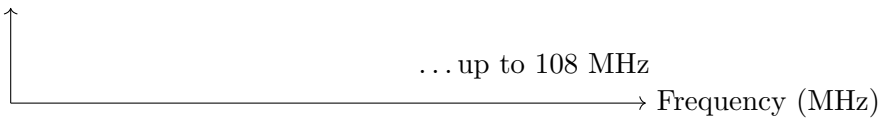
P5.11b: Find FM bandwidth for a 5-kHz voice signal with $\beta = 5$.

$$B_{\text{FM}} = 2(1+5) \times 5 = 2 \times 6 \times 5 = \mathbf{60} \text{ kHz}$$

P5.12b: Total number of FM channels in the FCC-allocated band.

$$\text{Channels} = \frac{108 - 88}{0.2} = \frac{20}{0.2} = \mathbf{100} \text{ channels}$$

(Of these 100, only 50 can operate simultaneously in a given area — alternating channels are kept silent to prevent interference.)



FCC FM Band: 88–108 MHz, 200-kHz channels, alternate channels used

2.3 Phase Modulation (PM) — 5.2.3

How PM Works

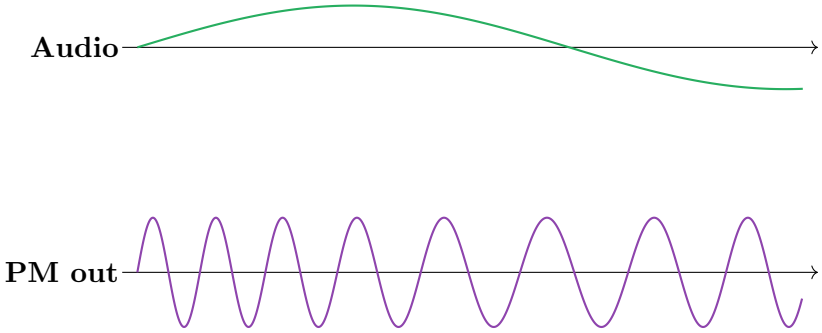
In PM:

- The **phase** of the carrier varies with the *amplitude* of the audio.
- Amplitude and frequency of the carrier remain *constant* (in theory).
- The key difference from FM: PM responds to the *amplitude* of the modulating signal; FM responds to the *integral* of it.
- Equivalently: PM is FM where the modulating signal is first passed through a **differentiator** (d/dt).

$$s(t) = A_c \cos\big(2\pi f_c t + k_p\, m(t)\big)$$

where k_p is the phase sensitivity constant.

Implementation: VCO + differentiator. The differentiator converts the audio into a derivative signal, which then drives the VCO exactly as in FM.



PM: phase shifts proportional to *instantaneous amplitude* of audio

2.3.1 PM Bandwidth

PM Bandwidth Formula

$$B_{\text{PM}} = 2\,(1+\beta)\,B_{\text{audio}}$$

- Formula is the same as FM, but β is typically smaller.
- Narrowband PM: $\beta \approx 1$.
- Wideband PM: $\beta \approx 3$.

Problem P5.11c — PM Bandwidth for 5-kHz voice, $\beta = 1$

$$B_{\text{PM}} = 2(1+1) \times 5 \text{ kHz} = 2 \times 2 \times 5 = \mathbf{20} \text{ kHz}$$

Significantly narrower than FM ($\beta = 5$ gives 60 kHz).

2.3.2 FM vs PM: The Key Difference

Property	FM	PM
What changes	Carrier frequency	Carrier phase
Controlled by	Amplitude of audio	Amplitude of audio
Mathematical relationship	$\Delta f \propto m(t)$	$\Delta\phi \propto m(t)$
Equivalently	PM of $\int m(t)\,dt$	FM of $\frac{d}{dt}m(t)$
Typical β	4 (broadcast)	1–3
Noise immunity	Very good	Good
Use	Broadcast radio, audio	Some digital systems

Master Summary — Chapter 5

Section 5.1 — Digital-to-Analog Conversion

Key formula: $S = N/r$, where $r = \log_2 L$, L = signal levels.

Method	Changes	BW	Noise	Use
BASK (OOK)	Amplitude	$(1+d)S$	Low	Optical, simple links
BFSK	Frequency	$(1+d)S + 2\Delta_f$	Medium	Modems, radio
BPSK	Phase	$(1+d)S$	Good	Base for QPSK
QPSK	Phase ($r = 2$)	$(1+d)S/2$	Good	Wi-Fi, satellite
16-QAM	Amp+Phase ($r = 4$)	$(1+d)S/4$	Better	Cable, DSL
64-QAM	Amp+Phase ($r = 6$)	$(1+d)S/6$	High	Cable modems

Constellation diagram: X-axis = I component, Y-axis = Q component. Distance from origin = amplitude; angle = phase. More points ⇒ more bits/baud but closer spacing ⇒ needs better SNR.

Section 5.2 — Analog-to-Analog Conversion

Method	Changes	BW formula	Real example
AM	Carrier amplitude	$2B_{\text{audio}}$	AM radio: 10 kHz, 530–1700 kHz
FM	Carrier frequency	$2(1+\beta)B_{\text{audio}}$	FM radio: 200 kHz, 88–108 MHz
PM	Carrier phase	$2(1+\beta)B_{\text{audio}}$	Narrowband: $\beta \approx 1$

AM radio: $B = 10$ kHz/station; $\frac{1170 \text{ kHz}}{10} = 117$ channels.
FM radio: $B = 200$ kHz/station; $\frac{20 \text{ MHz}}{0.2} = 100$ channels (50 active at once).
FM vs PM: Both use VCO; FM driven by $m(t)$, PM driven by $\frac{d}{dt}m(t)$.